History of A.I.

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Overview

Classical A.I.

First A.I. Winter

Knowledge-based A.I.

Second A.I. Winter

Data-driven A.I.

Third A.I. Winter

Classical A.I.

IBM
IBM
IBM
IBM
IBM
TAPE RACK 1
US
WHITE OWL

Advances in Classic A.I.

Computational search

Computational logic

Natural language

Successes of Classic A.I.

General Problem Solver

ELIZA

SHRDLU

First A.I. Winter

Limitations of Classic A.I.

Limited computer power

Computational complexity

Symbolic computing limitations

Failures of Classical A.I.

Machine translation deficiencies

Perceptron limitations

Lighthill report

First A.I. Winter
1974 – 1980

Knowledge-based A.I.

ACCES

Advances in Knowledge-based A.I.

Expert system	Knowledge base	Inference engine

Successes of Knowledge-based A.I.

MYCIN
diagnoses diseases

XCON
manages inventory

AARON
creates paintings

Second A.I. Winter

Limitations of Knowledge-based A.I.

Expensive to maintain

Rigid and inflexible

Not very efficient

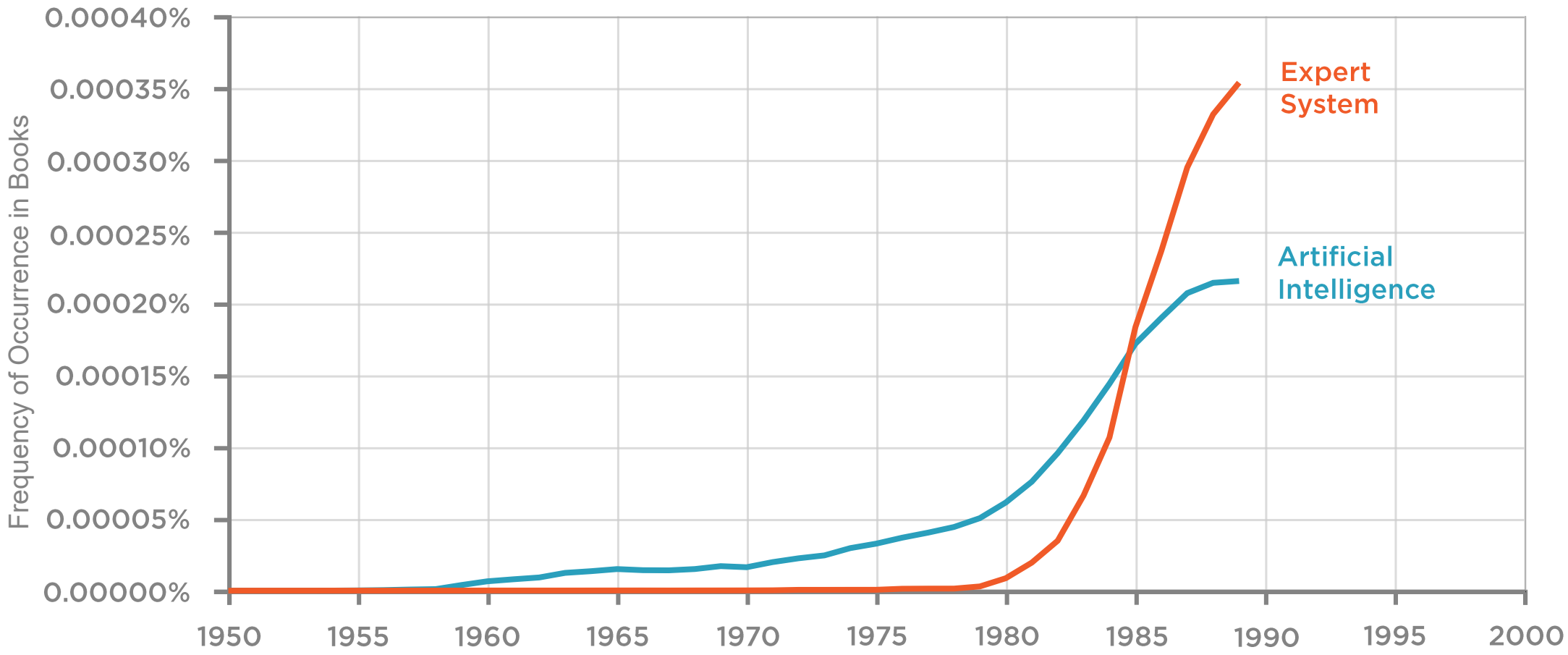
Failures of Knowledge-based A.I.

MYCIN never reaches production

Collapse of the LISP machine market

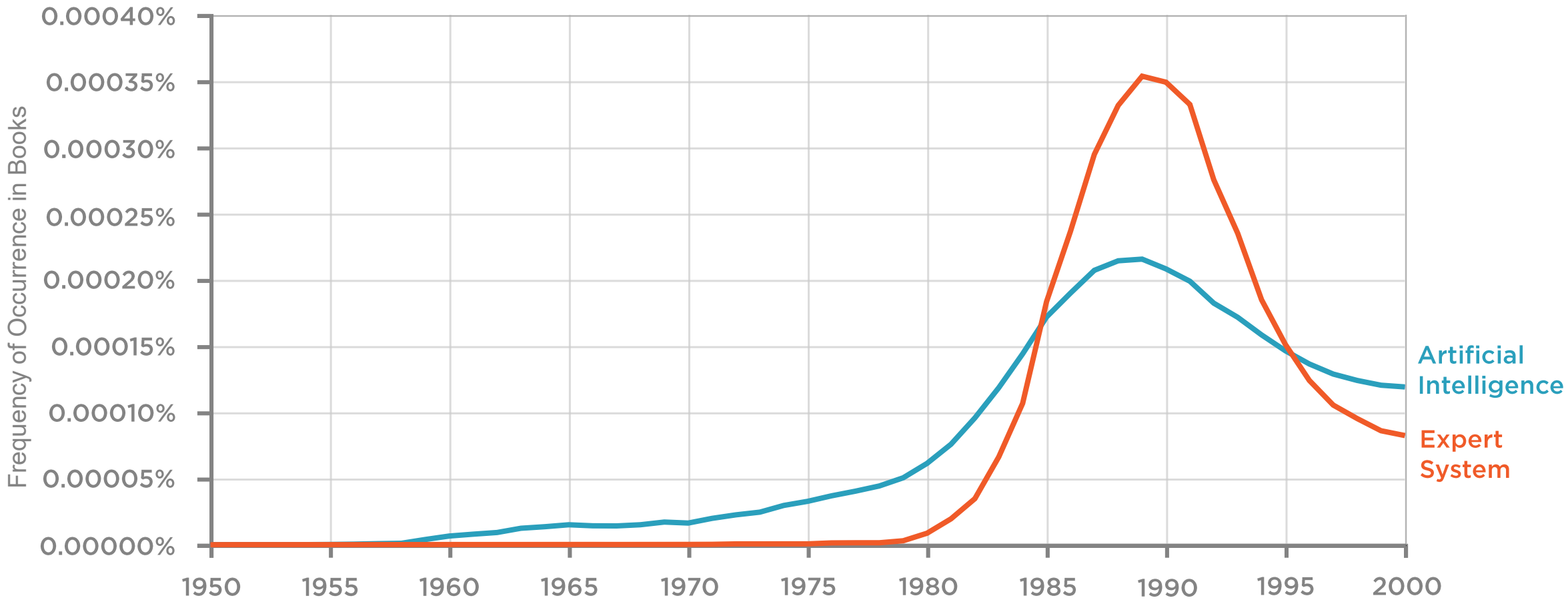
5th Generation Computer Systems Project

Second A.I. Winter



Source: Google Books Ngram Viewer

Second A.I. Winter



Source: Google Books Ngram Viewer

Second A.I. Winter
1988 – 2000s

Data-driven A.I.

A 3

Advances in Data-driven A.I.

More compute

More data

Better algorithms

More discipline

Successes in Data-driven A.I.

Stanley completes DARPA GC

Watson wins Jeopardy

AlphaGo wins at Go

Third A.I. Winter

Limitations of Data-driven A.I.

Limited reasoning capabilities

Requires labeled data

Only learns what you teach it

Failures of Data-driven A.I.

Self-driving vehicle collisions

Chat bot learns racism

Watson fails as a doctor

A Third A.I. Winter?

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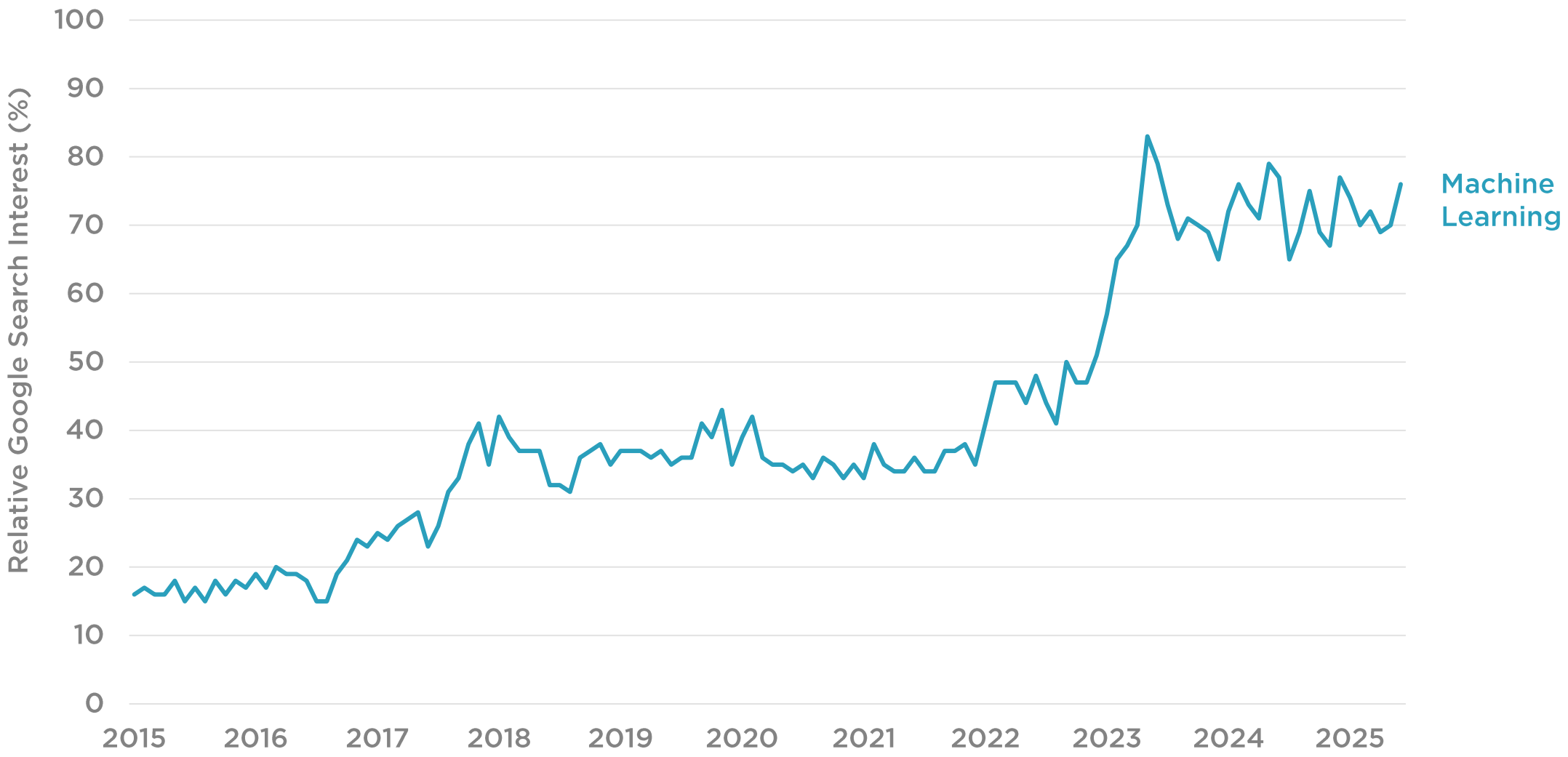
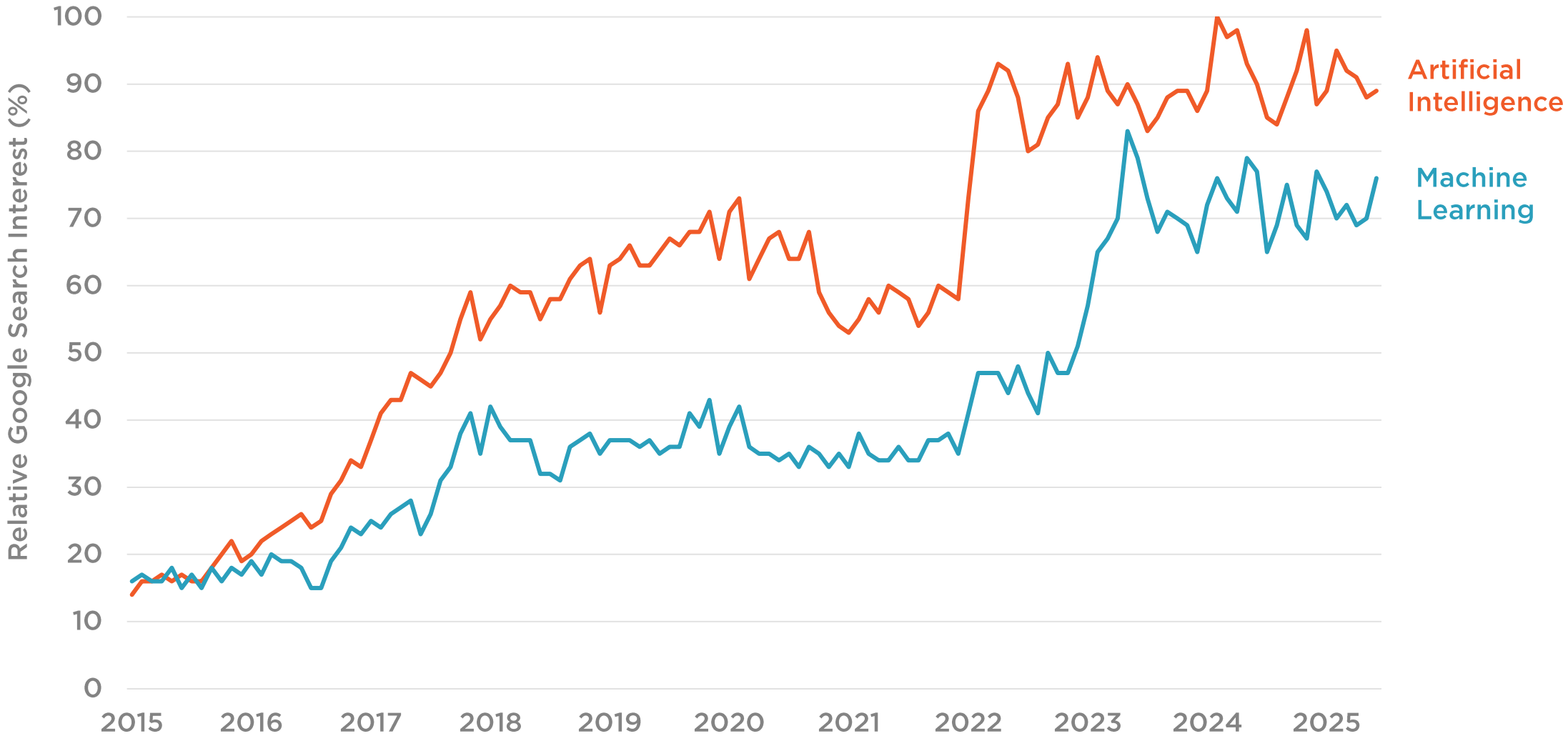


Source: Google Trends

A Third A.I. Winter?



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